Applied Catalysis A, General 607 (2020) 117861

Contents lists available at ScienceDirect

Applied Catalysis A, General

journal homepage: www.elsevier.com/locate/apcata

Effects of promoters (Mn, Mg, Co and Ni) on the Fischer-Tropsch activity and selectivity of KCuFe/mesoporous-alumina catalyst

Sandeep Badoga, Girish Kamath, Ajay Dalai*

Catalysis and Chemical Reaction Engineering Laboratories, Department of Chemical and Biological Engineering, University of Saskatchewan, Saskatoon, SK, S7N 5A9, Canada

ARTICLE INFO

Keywords:
Promoters Mn, Mg, Co, Ni
Fe K-edge XANES
XPS
Mesoporous alumina
Iron catalyst
Fischer-Tropsch synthesis

ABSTRACT

Fischer–Tropsch synthesis (FTS) is one of the promising technologies to produce liquid fuels and fine chemicals from coal, natural gas and biomass. Significant improvements in reactor technology and catalyst development are the focus of research in this area. In this work, the effects of 1.0 wt% promotor such as Mn, Mg, Co and Ni on the FTS activity and selectivity of mesoporous alumina supported KCuFe catalyst were studied. All synthesized promoted catalysts were characterized using BET, XRD, H_2-TPR, XPS, TEM and XANES. The FTS reaction was carried out in a fixed bed reactor at 270 °C, 300 psi, 2000 h^{-1} and $H_2/CO = 1.25$. In this work, the Ni-promoted catalyst showed the highest methane selectivity (~29 %). The Co-promoted catalyst showed the least Olefins/Paraffins (O/P) ratio and a ~5% increase in methane selectivity. Mg-promoted catalyst performed average but better than Ni and Co-promoted catalysts. However, Mn-supported catalyst outperformed Mg, Ni and Co-promoted catalysts and resulted in higher C_5+ selectivity along with a marginal increase in CO conversion. The ~5%, 2%, and 2% decline in CO_2, CH_4 and C_2-C_4 selectivities were also registered upon the addition of Mn promotor. The improvement in FTS activity and selectivity of Mn-promoted catalyst is related to the increase in iron dispersion and reducibility, as seen during H_2-TPR, XPS and XANES analyses. The presence of Mn^{+3} and Mn^{+4}, as confirmed by Mn 2p XPS and Mn K-edge XANES, indicates that the interaction at the electronic level between manganese and iron has contributed to improved performance. This study concludes that Mn is the most effective promotor for the synthesized mesoporous alumina supported KCuFe catalyst for the FT synthesis process.

1. Introduction

The utilization of biomass to produce clean fuels and chemicals is the need of today to contribute towards the reduction in greenhouse gas emissions. One of the pathways is utilizing the syngas ($CO + H_2$) produced from the gasification of biomass for Fischer–Tropsch synthesis (FTS) to produce liquid hydrocarbons. The FTS technology was invented in Germany and was extensively used during world war II to meet the fuel demands by German. Since then, various successful modifications in types of reactors, including stirred-tank, fixed bed, moving catalyst bed, and microchannel reactors [1] has been studied. A variety of heterogeneous catalysts, including supported and unsupported Fe and Co catalysts promoted with Pt, Ru, Ir, Re, Ag, K, Cu and Pd, have been tested, and some of them have shown improved performance [2,3]. Nevertheless, the research is still active in this area to achieve further higher and stable syngas conversions. The most active metals for the FTS process include Ru, Fe, and Co. Ru is the most active metal of all; however, it is ~225 times costlier than Co [4]; therefore, Fe and Co are preferred active metals. The selection between Co and Fe is typically made based on the quality of syngas. The syngas derived from methane/natural gas have higher H_2/CO ratios (> 2.0); therefore, the Co catalyst is preferred as it shows higher selectivity to paraffin, less water-gas shift activity and lower deactivation rates in comparison to the iron catalyst.

The Fe catalyst shows higher activity for water-gas shift reaction; therefore, it is preferred with coal or biomass-derived syngas where H_2/CO ratios are less than 2.0 (~0.5 − 2.0). The Fe catalyst thus manages the H_2/CO ratio via generating hydrogen through the water-gas shift, although in the process, it does convert CO to undesired CO_2. Furthermore, the coal-derived syngas contains impurities such as sulfur compounds, and the iron catalyst has higher tolerance as compared to Co [5].

The iron catalysts were found to produce lower molecular weight hydrocarbons, oxygenates and olefins. However, to increase the chain

* Corresponding author.
E-mail address: ajay.dalai@usask.ca (A. Dalai).

https://doi.org/10.1016/j.apcata.2020.117861
Received 7 February 2020; Received in revised form 20 September 2020; Accepted 22 September 2020
Available online 24 September 2020

growth probability α, potassium has been proven to be the best promotor. At higher potassium loading, α of $0.9-0.95$ was achieved by various researchers [5,6]. Copper is often used as a promoter with Fe catalyst, and it is believed to enhance the iron reduction during activation, thereby increasing the activity [7]. Typically, iron catalysts are unsupported; nevertheless, the research on the supported iron catalyst is active to address the issues such as catalyst separation from products [8]. In our recent work [9], mesoporous alumina and titania–alumina supports were synthesized and utilized to prepare KCuFe supported catalysts. The catalysts showed ~17 % higher CO conversions in comparison to corresponding γ-Al_2O_3 supported catalysts. Further, the effects of 0–12.5 % CH_4 and 0–15 % CO_2 in syngas on catalytic activity were also studied. The H_2/CO ratio of 1.25 was used for the FTS experiments in a continuous fixed bed reactor. Although the catalyst performance in terms of CO conversions and tolerance to impurities (CO_2 and CH_4) showed the potential for commercial application, however, further improvement in lowering the CO_2 selectivity and increasing C_5+ selectivity will add value to the process.

Therefore, four promoter's magnesium (Mg), cobalt (Co), manganese (Mn) and nickel (Ni), were shortlisted to study their effects on the catalyst performance. Mg is known to help in the suppression of CH_4 and inhibits the WGS activity [10,11]. Mg also leads to form smaller Fe_2O_3 crystallites and enhances the reduction and carburization. Moreover, it helps in increasing olefins selectivity [10–13] and shifts the hydrocarbon production to the gasoline range.

Monte et al. [14] studied the promotion effect of cobalt on the iron catalysts. They observed a synergistic effect of K and Co on Fe catalyst and reported the increase in activity, α and stability. A significant amount of research conducted on Co-Fe bimetallic catalysts showed an increase in dispersion, reducibility, CO conversion, C_5+ selectivity and α. [15,16]. Furthermore, the decrease in CH_4 and CO_2 selectivities was also observed. However, limited literature is available on the use of Co as a promoter on Fe catalyst.

Nickel, as a promoter, increases the iron oxide dispersion, thus reduces the crystallize size. Moreover, it provides a higher rate of reduction and carburization in H_2 and CO, respectively. It is known to stabilize the catalyst and increase the formation of shorter hydrocarbons. However, it has an inherent tendency for the formation of methane [17–20]. Therefore, Ni acting as a promoter might increase the CO conversion.

Mn is a proven promoter to reduce the formation of CH_4 selectivity and increase the olefins and C_5+ selectivity, and to optimize the distribution of the product. Mn is an electron-donating promoter, thus facilitates dissociative adsorption of CO, which leads to an increase of chain growth probability, α hence heavy hydrocarbons [21,22]. Mn in small quantity (Mn/Fe weight ratio of 0.06) was found to enhance the CO conversion and decrease the deactivation rate. Mn tends to increase iron oxide dispersion and also contributes to the reduction and carburization [23–25]. Along with electronics, Mn also acts as a structural promoter. It stabilizes the catalytic activity by retaining a high surface area of iron carbides. Fe and Mn (comparable ionic radii) tend to form mixed oxides, which inhibit crystallite growth and thus retained smaller Fe particles.

Therefore, in this work, Ni, Mg, Mn and Co promoted KCuFe catalysts supported on mesoporous alumina were synthesized and tested for Fischer-Tropsch synthesis (FTS) in a fixed bed reactor. The catalysts were characterized using BET, XRD, H_2-TPR, XPS, and XANES. The FTS reaction was carried out at 270 °C, 300 psi, 2000 h^{-1} and H_2/CO = 1.25. The products of the reaction are analyzed using online gas-GC and simulated distillation. The effects of a promoter (Ni, Mg, Mn, Co) on catalytic activity and Fischer-Tropsch product optimization were studied in detail.

2. Experimental

2.1. Materials

Pluronic P123 polymer, aluminum isopropoxide, iron (III) nitrate nonahydrate, cupric nitrate trihydrate, potassium nitrate, cobalt (II) nitrate hexahydrate, nickel nitrate hexahydrate, manganese nitrate tetrahydrate and magnesium nitrate hexahydrate were purchased from Sigma Aldrich, Canada. Anhydrous ethanol and HNO_3 were purchased from Fischer-Scientific, Canada. Mixed syngas 40 % CO, 10 % Ar and balance H_2, and ultrapure H_2 was purchased from Praxair, Canada.

2.2. Catalyst synthesis

Mesoporous alumina was synthesized using the procedure mentioned in our previous work [26]. The catalysts were synthesized using the sequential incipient wetness impregnation method. A respective amount of iron precursor resulting in 25 wt % of Fe was dissolved in ethanol/water mixture and loaded on mesoporous alumina. After drying, the material at 120 °C, 0.5 wt % of Cu was wet impregnated. The resulting material was dried and then calcined at 400 °C for 4 h at a ramp rate of 1 °C/min. To this material, 1.0 wt % K was wet, impregnated, dried and calcined. The resulting catalyst is named as KCuFe/mAl_2O_3. 1.0 wt% promoter (Ni, Co, Mn, Mg) was then impregnated to KCuFe/mAl_2O_3. The material was dried and then calcined at 400 °C for 4 h at a ramp rate of 1 °C/min to obtain catalysts Co-KCuFe/mAl_2O_3, Ni-KCuFe/mAl_2O_3, Mn-KCuFe/mAl_2O_3, and Mg-KCuFe/mAl_2O_3.

2.3. Material characterization

The surface area was determined using Brunauer–Emmett–Teller, BET method and pore size analysis was using Barrett–Joyner–Halenda, BJH method. ASAP2020 Micrometrics instrument was used for determining BET and BJH using N_2 adsorption-desorption. Wide-angle (2θ = 20° to 80°) powder X-ray diffractograms for each material was obtained using Bruker Advance D8 series II, Powder diffractometer with Cu Kα radiations. The temperature-programmed reduction (TPR) profile using hydrogen for the reduction of catalyst (iron) was obtained using Autochem 2950 HP Micromeritics instrument equipped with TCD detector. The instruments ASAP2020 Micrometrics, Bruker Advance D8 series II XRD and Micromeritics Autochem 2950 HP were housed in Catalysis and Chemical Reaction Engineering Laboratories (CCREL), Department of Chemical and Biological Engineering, University of Saskatchewan. The details of sample preparation and analyses have been described in our previous paper [27].

The X-ray Photoelectron Spectroscopy (XPS) was performed using a Kratos (Manchester, UK) AXIS Supra system, which is equipped with a 500 mm Rowland circle monochromated Al K-α (1486.6 eV) source and combined hemispherical analyzer (HSA) and spherical mirror analyzer (SMA). The survey scan spectra for Co, Ni and Mn containing materials were taken for binding energy BE $5-1200$ eV, whereas, for Mg containing material, the survey scan was taken for BE up to 1325 eV. The scan was done in 1 eV steps with a pass energy of 160 eV. High-resolution scans for Co2p, Fe2p, Mg 1 s, Mn2p and Ni 2p were conducted using 0.1 eV steps with a pass energy of 20 eV. The XPS system used for this work is at Saskatchewan Structural Sciences Centre (SSSC), University of Saskatchewan.

The X-ray absorption spectroscopy (XAS) on the catalysts was performed at Canadian Light Source, Saskatoon, Saskatchewan. All catalysts were reduced under hydrogen at 400 °C for 18 h before analysis. The samples for XAS analysis were dispersed on a conductive carbon tape in a glove box under helium. The samples were analyzed at Soft X-Ray Microcharacterization Beamline (SXRMB) beamline. The Fe K-edge absorption spectra were obtained in total electron yield. The data was analyzed using Athena software.

2.4. Fischer–Tropsch experimental details

The Fischer–Tropsch synthesis (FTS) was carried out in a continuous fixed-bed tubular reactor. The experimental setup was described in our previous work [26]. The system consists of a set of mass flow controllers meant to control the gas flow rate of H_2 and syngas ($CO + H_2$) to the tubular reactor. The reactor is cased in a vertical furnace, which controls temperature via an automatic temperature controller. The product out of the reactor was channeled to the hot (~120 °C) gas-liquid separator, where heavy hydrocarbons get condensed, and the remaining gaseous product is routed to a cold condenser (~0 °C) to trap all the condensable hydrocarbons. The remaining gaseous products were directed to a vent, and a small stream was sent to an online GC. The entire setup was maintained under reaction pressure using a backpressure regulator. After the reaction, the liquid hydrocarbons were collected from the hot and cold trap.

The tubular reactor is 50 cm long and having an ID of 1 cm. 2 g of powder catalyst was diluted with 90 mesh silica carbide (SiC) and loaded in the reactor following the reactor packing described in previous work [27]. Before the FTS reaction, the catalyst was reduced in-situ at 380 °C for 16 h under hydrogen flowing at 30 mL/min. The reactor was then allowed to cool down to 200 °C, followed by pressurizing to 300 psi. The syngas (H_2/CO = 1.25) was then introduced at the flowrate to achieve a gas hourly space velocity of 2000 $h^{-1,}$ and the temperature was ramped to reaction temperature 270 °C with a ramp rate of 1 °C/min.

The product gas from the FTS system was analyzed every 12 h in online GC (Shimadzu-2014) equipped with FID and TCD detectors to determine the concentration of CO and specific hydrocarbons. For each analysis, three consecutive GC injections were made; an average of these was used to report the FTS activity and selectivity data. Among the three GC injection data, the error in measurement was less than ± 1.5 %. The analysis of product gas was also performed in another GC 7890A (Agilent GC) to get the concentration of some of the C_2-C_4 compounds, which online GC could not determine. The GC 7890A is equipped with two TCD and one FID detectors and has a complex system of five columns.

The liquid product hydrocarbons were separated from the water via centrifugation and phase separation. The hydrocarbon products were analyzed in the Varian CP-3800 GC instrument for boiling point distribution, which is then related to the carbon number. The GCs used for analyzing liquid and gas products are housed in CCREL, University of Saskatchewan, Canada.

3. Results and discussion

3.1. Textural properties

The textural properties, including pore volume, pore diameter and surface area of all catalysts, are shown in Table 1. The N_2 adsorption-desorption isotherms and the pore size distribution are shown in Fig. 1. The textural properties of mesoporous alumina support are also reported in Table 1, and the adsorption isotherms are shown in our previous work [26]. The synthesized mesoporous alumina and all catalysts represent Type IV isotherm. Based on the classification by IUPAC in 2015, the mesoporous alumina synthesized in this work resembles H1 type hysteresis loop [28,29], which is typical for ordered mesoporous material with cylindrical pores. However, on impregnating the support with KCuFe, the hysteresis loop type changed to H2(b), as seen in Fig. 1. Further on the addition of promoters, the hysteresis loop tends to be like type H2 (a). The type H2 hysteresis loop results from a disordered complex pore system, where pores can have narrow openings such as ‘ink-bottle’ pores. The type H2 (a) indicates that the “size distribution of pore cavities is relatively wide compared to the distribution of neck sizes,” whereas type H2 (b) is observed, “in the case of the narrow size distribution of pore cavities” [28].

The surface area, pore volume and pore diameter of Mesoporous alumina (mAl_2O_3) are 430 m^2/g, 0.9 cm^3/g and 7 nm, respectively (Table 1). On the addition of 26.5 wt % of metals (Fe, K and Cu), the surface area, pore volume and pore diameter are decreased to 344 m^2/g, 0.4 cm^3/g and 5 nm. The decrease in surface area is proportional to the metal loading. However, the decline in pore sizes after the impregnation of metals indicates a change in the pore structure. The change in the type of hysteresis loop from H1 to H2 on the addition of metal also indicates the same. Further decrease in textural properties on the addition of promotors is nearly proportional to the amount (1 wt %) added. The slight difference in surface area and pore diameter of catalysts with different promotors could be attributed to the difference in the extent of metal-metal interactions. In this study, Mn-KCuFe/mAl_2O_3 catalyst showed lesser surface area, pore volume and pore diameter of 305 m^2/g, 0.35 cm^3/g and 4.4 nm, respectively.

Table 1
Textural properties of all promoted catalysts, KCuFe/mAl_2O_3 catalyst and support material.

Material	Surface area (m^2/g) (± 3%)	Pore volume (cm^3/g)	Pore diameter (nm)
Mesoporous alumina (mAl_2O_3)	430	0.9	7.0
KCuFe/mAl_2O_3	344	0.4	5.0
Mn-KCuFe/mAl_2O_3	305	0.35	4.4
Ni-KCuFe/mAl_2O_3	329	0.4	4.6
Mg-KCuFe/mAl_2O_3	313	0.4	4.8
Co-KCuFe/mAl_2O_3	337	0.4	4.8

3.2. X-ray diffraction

In heterogeneous catalysis, XRD plays an important role in identifying the type of crystallographic structure of active metal(s) present in powder catalyst. It provides an estimation of the catalyst performance. In an oxide catalyst, the iron oxide can be present in hematite (Fe_2O_3) and magnetite (Fe_3O_4) forms. The peaks at 2θ = 33.2, 40.9, 49.7, 54, 57.6, 61, 64, 68 and 72 degrees, and 2θ = 35.4, 37, 43.4, 53.5, 57, 62.5, 71, 74 and 75 degrees in XRD profiles represent the presence of hematite and magnetite, respectively [30]. However, in reduced/activated catalysts, different phases can be observed: α-Fe, magnetite and iron carbide (χ-Fe_5C_2, ε-$Fe_{2.2}C$) [31]. Fig. 2 shows the XRD profiles for all catalysts. The absence of clear and sharp peaks implies a fine dispersion of iron oxide. The XRD profile for KCuFe/Meso-Al shows broad and small peaks at 2θ = 33.2, 35.4, 49.7, 53.5, 62, 72 degrees. However, on adding promotors, these peaks are not visible. This indicates that all of the promotors have aided in increasing the iron oxide dispersion.

3.3. Temperature programmed reduction

One of the anticipated roles of the promotors (Ni, Co, Mn and Mg) is easing the reduction of iron-oxide. However, the extent is dependent on the type and quantity of promoter used. Therefore, to understand the effect of promotors on the reducibility of iron-oxide, the hydrogen temperature-programmed reduction (H_2-TPR) study was performed. Fig. 3 shows the H_2-TPR profile for all promoted catalysts. The H_2-TPR profile for reference catalyst KCuFe/mAl_2O_3 is also shown in Fig. 3. Two peaks are evident in the H_2-TPR profile. The peak at a lower temperature (~290 – 350 °C) represents the reduction of Fe_2O_3 (Fe III) to Fe_3O_4 (Fe II, III), and the broad peak at a higher temperature (~400 – 500 °C) is due to the reduction of Fe_3O_4 to FeO (Fe II) followed by reduction of FeO to Fe.

The reduction peak at higher temperatures indicates the difficulty in reduction. The area under the curve is proportional to the amount of hydrogen uptake, which is proportional to the extent of reduction.

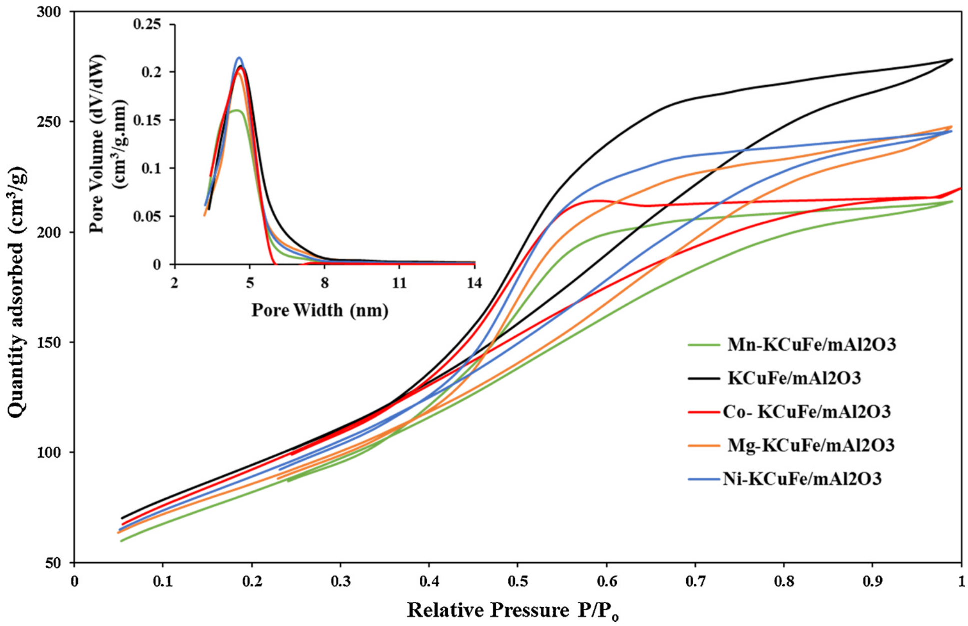

Fig. 1. N_2 adsorption-desorption isotherms and inset is the pore size distribution for all catalysts in this study.

Therefore, the higher hydrogen uptake and lower reduction temperature imply the ease in reducibility. Fig. 3 shows that the iron oxide reduction peaks for KCuFe/mAl_2O_3 catalyst appeared at 340 °C and 500 °C. As seen in the TPR profiles, the addition of promotes has further lowered the reduction temperatures. Mn, Mg, Ni, and Co-promoted catalysts have shown reduction peaks at 302 °C and 420 °C, 305 °C and 430 °C, 308 °C and 420 °C, and 314 °C and 430 °C, respectively. This confirms that Mn and Ni performed a little better than the Mg and Co in facilitating the iron oxide reduction. The hydrogen uptake for all catalysts is shown in Table 2. The Co-promoted catalyst shows the least amount of hydrogen uptake followed by the Mg-promoted catalyst. Ni and Mn promoted catalysts are similar in hydrogen uptake; nevertheless, they show a higher amount of hydrogen uptake in the second reduction peak than the reference catalyst KCuFe/mAl_2O_3. The H_2-TPR study proved that the greater extent of reduction could be achieved in promoted catalysts, specifically in Mn and Ni promoted catalysts and at a temperature ~ 70 °C lower than that in KCuFe/mAl_2O_3 catalyst.

3.4. X-ray photoelectron spectroscopy

The XPS analysis was performed to reveal the chemical states of active metal to understand the effects of promotors at the electronic level. The XPS spectra were calibrated for 1 s C peak at 284.8 eV. The Fe 2p XPS spectra for all promoted catalysts are shown in Fig. 4. The 2p peak splitting is due to spin-orbit coupling resulting in two peaks in the area ratio 1:2 as $2p_{1/2}$ and $2p_{3/2}$. The split peaks are at 13.1 eV energy difference. The binding energy (BE) for these peaks represents the chemical states. For Fe_2O_3, oxidation state Fe^{+3}, the $2p_{3/2}$ B.E is 710.8 eV, for FeO, oxidation state Fe^{+2} the $2p_{3/2}$ B.E is 709.6 eV and B.E for Fe° is 706.7 eV [32]. The lower BE for lower oxidation state is obvious as less energy is required to eject an electron from Fe^{+2} instead of Fe^{+3}. The $2p_{3/2}$ BE for Fe in all promoted catalyst is ~ 711 eV, which suggests the presence of Fe_2O_3 (Fe^{+3}). Additionally, the $2p_{3/2}$ satellite feature at ~719 eV ($\Delta eV \approx 8$) and the $2p_{1/2}$ satellite feature at ~733 eV are a strong indicator of Fe^{+3}.

It can be noted that the binding energy for Fe $2p_{3/2}$ peak is slightly different for different promotors. Additionally, the area under the curve is also different for different promoters. This could be associated with the electron transfer between the promotor and iron oxide. The decrease in 2p peak area or peak intensity indicates lesser iron concentration on the surface of the catalyst [33]. The atomic percentage of iron on the surface of catalysts was calculated using CasaXPS demo

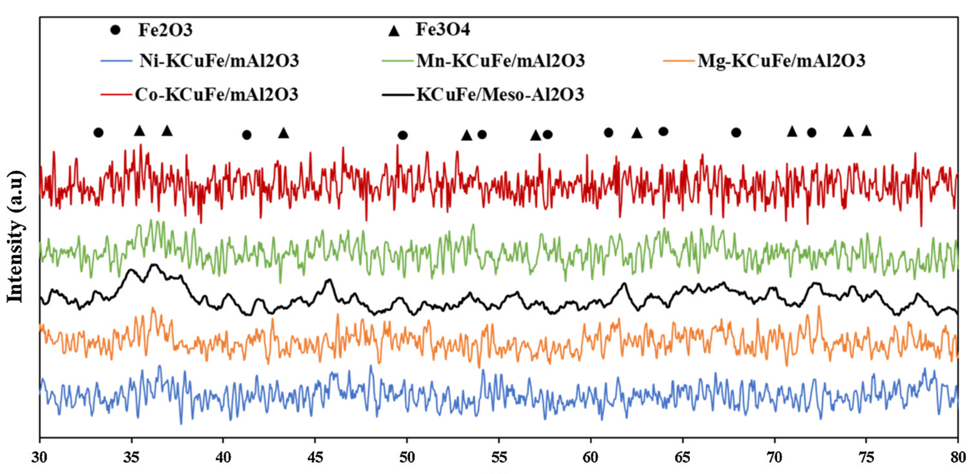

Fig. 2. XRD profile for all catalysts in this study.

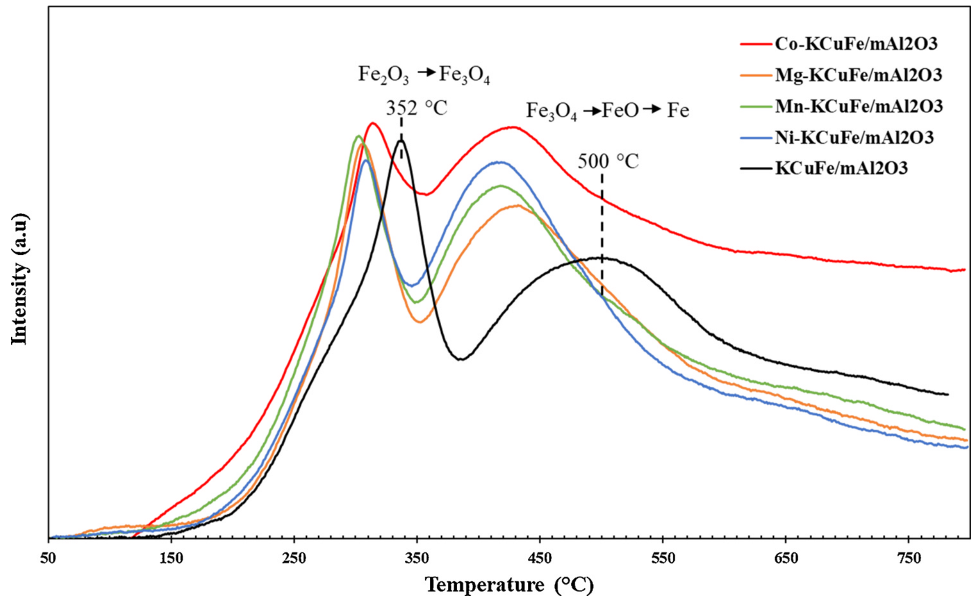

Fig. 3. Hydrogen- Temperature program reduction profile for all catalysts in this study.

Table 2
Hydrogen uptake by catalysts during reduction.

Catalysts	Quantity of H_2-uptake (μmol/g) (± 5%)	
	Low temperature peak	High temperature peak
KCuFe/mAl_2O_3	26	30
Mn-KCuFe/mAl_2O_3	26	32
Ni-KCuFe/mAl_2O_3	24	35
Mg-KCuFe/mAl_2O_3	23	27
Co-KCuFe/mAl_2O_3	19	23

Table 3
The atomic percentage of iron on the surface of oxide promoted catalysts.

Catalyst	Fe atomic % (± 10 %)
Co-KCuFe/mAl_2O_3	6.0
Mg-KCuFe/mAl_2O_3	4.5
Mn-KCuFe/mAl_2O_3	8.0
Ni-KCuFe/mAl_2O_3	6.5

software and shown in Table 3. Among all promoted catalysts, the Mn-promoted catalyst showed the highest atomic percentage of iron on the surface. The promoters follow the order Mn > Ni > Co > Mg for the concentration of iron at the catalyst surface. Mg-promoted catalyst has almost 40 percent less iron on the surface than that in Mn-promoted catalyst.

The XPS spectra for each promoter in their corresponding catalysts are also shown in Fig. 5. Mg 1 s XPS peak appearing at 1304 eV and oxygen 1 s peak (not shown in the figure) appearing at ~531 eV confirm the presence of MgO [34]. Co $2p_{3/2}$ XPS peak in cobalt oxide is typically at ~780 eV; however, the Co 2p XPS in Co-promoted catalyst

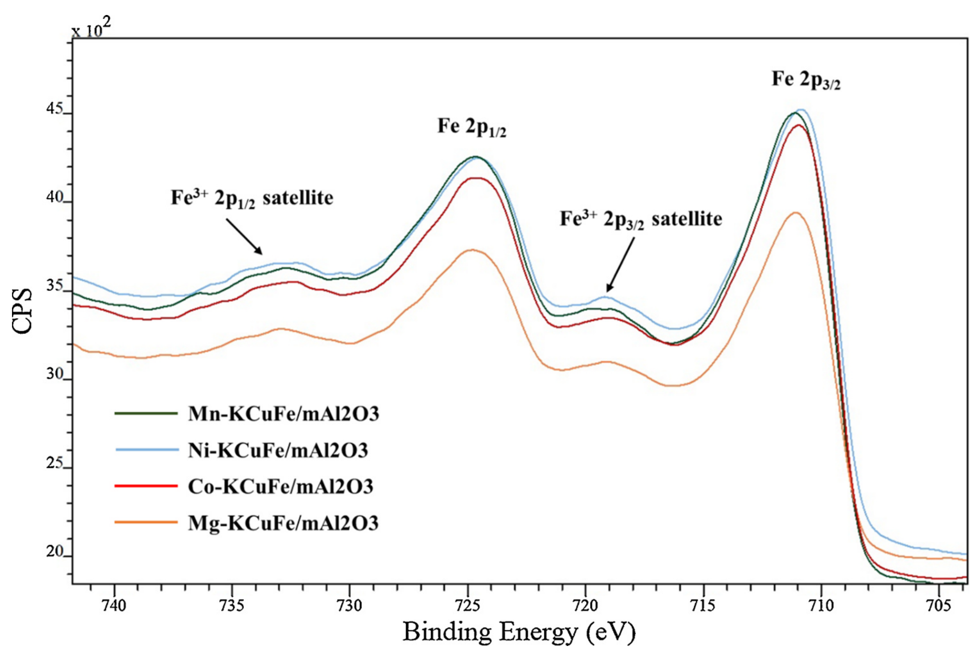

Fig. 4. Fe 2p XPS spectra for oxide promoted catalysts.

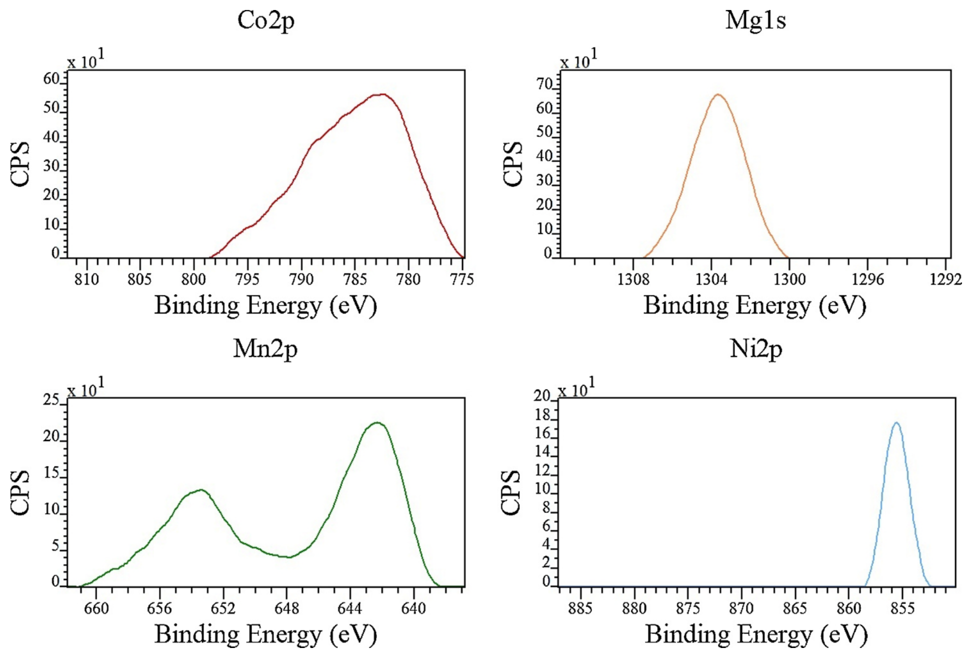

Fig. 5. XPS spectra of Co, Mn, Mg and Ni present in corresponding catalysts.

is at 782.5 eV. This indicates the presence of cobalt aluminate [35] or strong interaction of Co and Fe, forming Co-Fe bimetallic system. Mn 2p XPS spectra shows a split peak $2p_{1/2}$ and $2p_{3/2}$ at ~653.9 eV and ~642.5 eV. Based on literature database, the $2p_{3/2}$ XPS peak for Mn^{2+} is at 641.5 eV [32]. The higher BE of Mn in Mn-prompted catalyst indicates presence of Mn^{+3} and Mn^{+4} oxidation states. This could be due to the interaction between Mn and Fe. Ni $2p_{3/2}$ XPS peak in nickel oxide appears at 854 eV; however, for Ni promoted catalyst, it appeared at 855.7 eV. This indicates the formation of nickel aluminate [36,37] or a strong bond with hydroxides on the surface of alumina.

3.5. X-ray absorption spectroscopy

The Fe K-edge X-ray absorption near-edge spectra (XANES) for all promoted reduced catalysts are shown in Fig. 6. The intense peak at ~7135 eV is due to 1 s to 4p electronic transition. The Fe K-edge resembles the K-edge of Fe Foil [38], indicating the reduction of iron. It was observed that the binding energy (K-edge) for each promoted catalyst is slightly different. Lower BE implies a higher reduction; therefore, based on Fig. 6, the order at which the promotors enhanced the reduction of iron is Ni > Mn > Co ≥ Mg. Further, Fig. 7 shows the R-space EXAFS for Fe K edge of reduced catalysts. The first peak represents the first shell of Fe surrounding the Fe. The R-space is similar to that observed in Fe foil [39,40], indicating the reduction of iron in all catalysts. It can be seen that for reduced catalysts containing different promotors, the R-space between first, second, third and fourth shell is different. This implies that different promotors have a different electronic influence on the active metal, iron. To confirm the interaction between Mn and Fe, Mn K-edge XANES spectra of reduced Mn-KCuFe/mAl_2O_3 catalyst was recorded (see Fig. 8). The Mn K-edge absorption energy is between the K-edge absorption energy for metallic Mn and the MnO [41,42]. This suggests that Mn is present in the catalyst in the form of oxygen-deficient MnO [42], which could also be possible if strong interaction between Mn and Fe is present. Therefore, XANES studies confirmed that different promotors interact with Fe electronically, and the extent is dependent on the type of promotor.

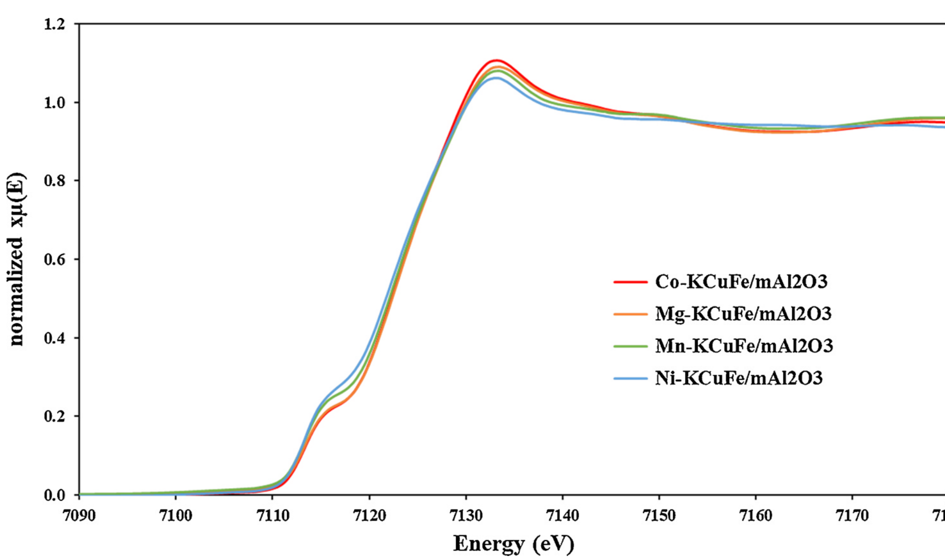

Fig. 6. Fe K-edge XANES spectra for reduced promoted catalysts.

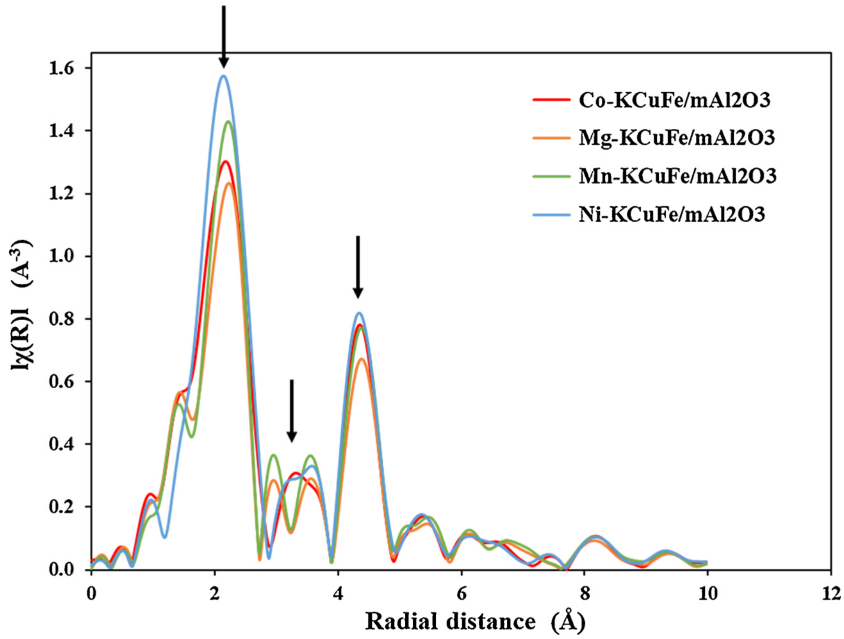

Fig. 7. Fe K-edge EXAFS spectra in R-space for reduced catalysts.

3.6. Catalyst performance for Fischer–Tropsch synthesis

The FTS reactions were carried out in a fixed bed reactor, and each catalyst was tested at the same operating conditions: 270 °C, 300 psi, H_2/CO = 1.25, and GHSV = 2000 h^{-1}. The catalytic activities for all catalysts in terms of CO conversion, and CH_4, CO_2, C_2-C_4 and C_5+ selectivity are reported in Table 4. The unpromoted catalyst Fe/mAl_2O_3 showed higher CO conversion (94 %); however, the majority of the CO was converted to CO_2 (~46.9 %). Also, among hydrocarbons, the CH_4 selectivity was 34.1 %, and the C_5+ selectivity was measured to be around 55 %. The addition of K and Cu promoters has resulted in significantly lowering the methane selectivity to 12 %, whereby increasing C_5+ selectivity to 77 %. The CO_2 selectivity was also decreased by ~3%.

The Co-KCuFe/mAl_2O_3 catalyst showed similar CO conversion but higher CH_4 selectivity and least olefin/paraffin (O/P) ratio compared to KCuFe/mAl_2O_3 catalyst. The decrease in the extent of active metal reduction in Co-prompted catalyst as observed in H_2-TPR analysis and Fe K-edge XANES analysis could be the reason for its lower performance. The XPS analysis of oxide catalysts also indicated the presence of cobalt aluminate (which might have been formed due to the introduction of Co(II) in the tetrahedral structure of mesoporous alumina framework) [43]. The presence of cobalt aluminate has been observed in low cobalt loading in various works [44,45]. Therefore, in our work, the formation of Co–Fe system could possibly explain ~ 5% increase in methane selectivity [46] as compared to the KCuFe/mAl_2O_3 catalyst. It leads to a decline in C_5+ selectivity. The C_2-C_4 selectivity was unaffected on the addition of cobalt as a promoter.

The Mg-KCuFe/mAl_2O_3 catalyst showed similar CO conversion as compared to that shown by Co-KCuFe/mAl_2O_3 catalyst. The determination of the atomic percentage of iron on the surface of the catalyst using XPS survey spectra indicated that Mg- promoted catalyst has the least percentage of iron on the surface as compared to other promoted catalysts. This could be the reason behind the reduction in O/P ratio

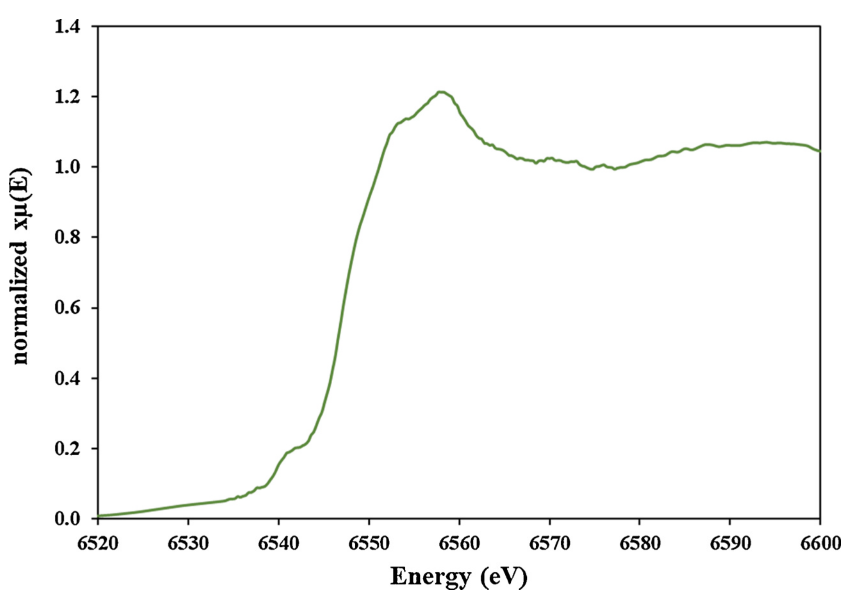

Fig. 8. Mn K-edge XANES spectra for reduced Mn-KCuFe/mAl_2O_3 catalyst.

Table 4
Catalyst performance data for FTS (Reaction conditions: 270 °C, 300 psi, H_2/CO = 1.25 and GHSV = 2000 h^{-1} (TOS = 70 h)).

Catalysts	CO Conversion, %	Selectivity, %				Olefin/Paraffin ratio
		CO_2	CH_4	C_2-C_4	C_5+	
Co-KCuFe/mAl_2O_3	93	43.2	16.8	11.0	72.2	0.25
Ni-KCuFe/mAl_2O_3	77	40.7	29.3	12.6	58.1	0.30
Mg-KCuFe/mAl_2O_3	93	42.9	14.8	11.2	74.0	0.60
Mn-KCuFe/mAl_2O_3	95	39.4	10.0	9.0	81.0	0.85
KCuFe/mAl_2O_3	94	44.0	12.0	11.0	77.0	0.71
Fe/mAl_2O_3	95	46.9	34.1	11.1	54.8	0.35
0.5Mn-Fe/mAl_2O_3	95	43.3	25.3	10.7	64.0	0.56
1Mn-Fe/mAl_2O_3	94	45	26.1	10.9	63.0	0.55
2Mn-Fe/mAl_2O_3	94	45.5	25.6	10.8	63.6	0.54

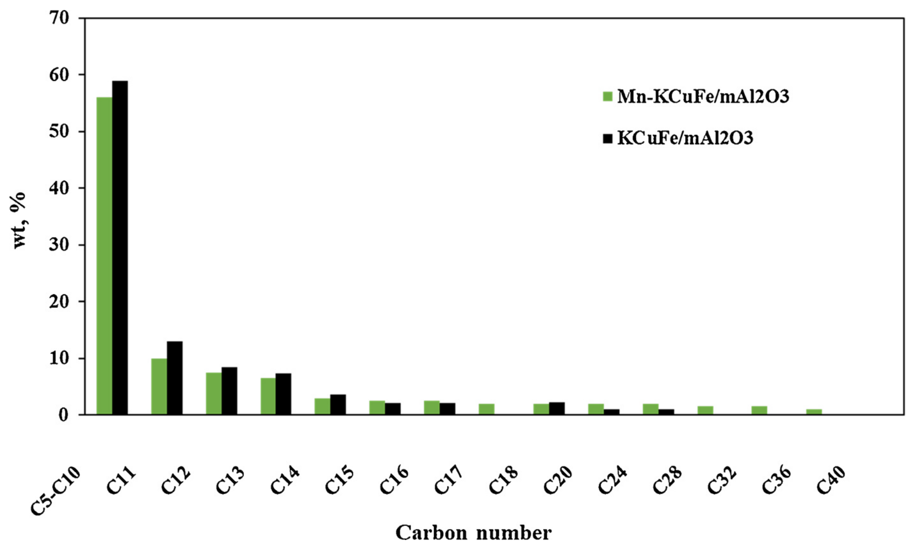

Fig. 9. Boiling point distribution of liquid hydrocarbon products collected from FTS reaction over Mn-KCuFe/mAl_2O_3 and KCuFe/mAl_2O_3 catalysts.

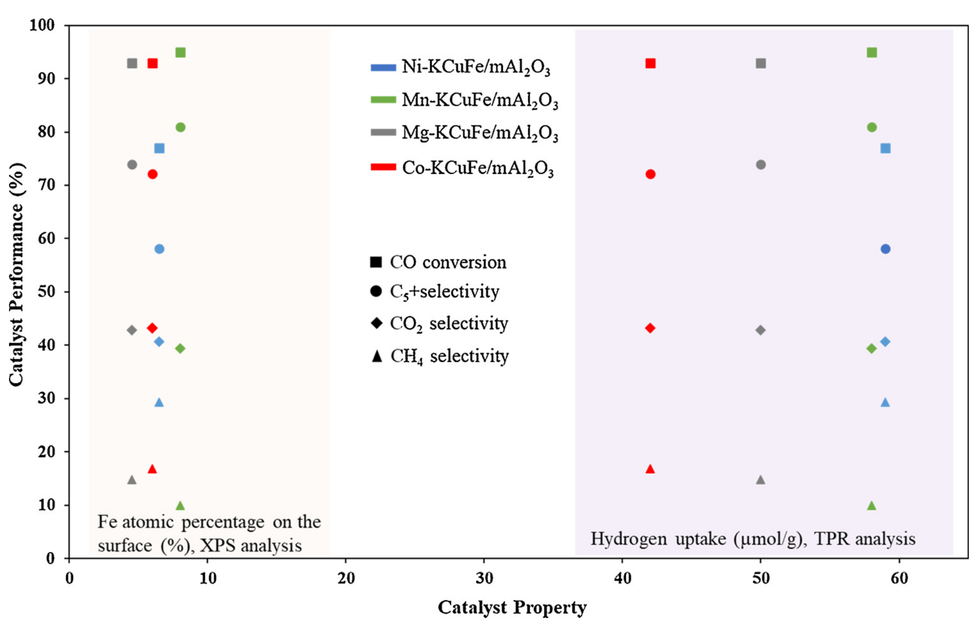

Fig. 10. Correlation between catalysts properties and performance.

from 0.71 to 0.60 and increase in CH_4 selectivity from 12.0 % to 14.8 % as compared to KCuFe/mAl_2O_3 catalyst. Similar CO_2 selectivity implies that Mg did not play a significant role in reducing WGS activity. The Fe K-edge XANES analysis of reduced catalysts confirmed that Mg-promoted catalyst has a slightly lower extent of reduction as compared to Mn and Ni promoted catalysts. The Ni-KCuFe/mAl_2O_3 catalyst showed the highest extent of active metal reduction as evident from H_2-TPR and XANES. However, this catalyst showed 18 % less CO conversion as compared to the unpromoted catalyst. Further, this catalyst showed the highest methane selectivity of 29.3 % among all catalysts, probably due to the hydrogen spillover potential of nickel and due to Fe–Ni interactions as seen in the Ni 2p XPS spectra. The decrease in CO_2 selectivity by ~2.5 % implies reduction in WGS activity. Decline in WGS activity, lower H_2/CO ratio, and high methane selectivity might have created the hydrogen deficiency, resulting in the lowest CO conversion among all catalysts.

The Mn-KCuFe/mAl_2O_3 catalyst is the only one among promoted catalysts, which performed better than other catalysts. The highest atomic percentage of iron on the surface of the catalyst (XPS analysis) and aid in easing the iron reduction (H_2-TPR analysis) are contributing to its best performance among all catalysts. The Mn 2p XPS spectra showing the presence of Mn^{+3} and Mn^{+4}, and analysis of Mn K-edge XANES spectra indicates the electron-donating tendency of Mn and implies the electronic interactions between Mn and Fe. This synergy between Mn and Fe might have played a significant role in improving the catalytic performance. To further elucidate the role of Mn promotion, three catalysts with 0.5 wt % Mn, 1.0 wt % Mn and 2 wt % Mn on Fe/mAl_2O_3 were synthesized and tested for FTS under similar reaction conditions (see Table 4). It was observed that on the addition of 0.5 wt % Mn, there was a significant decrease in methane selectivity and an increase in C_5+ selectivity. However, with increase in the amount of Mn from 0.5 to 2 wt %, there was not much positive effect observed.

Therefore, with the addition of Mn on KCuFe/mAl_2O_3 catalyst, the decrease in methane, CO_2 and C_2–C_4 selectivity and thus increase in C_5+ selectivity to 81 % from 77 % was observed (Table 4) as expected from the isolated promotor study discussed above. Further, the O/P ratio also increased to 0.85 from 0.71. Therefore, this study concludes that Mn is the best among all promotors studied in this work, and it is capable of lowering the CO_2 selectivity and increasing the much desired C_5+ selectivity. Further, the boiling point distribution of the liquid hydrocarbon products collected from FTS over Mn-KCuFe/mAl_2O_3 catalyst was performed and shown in Fig. 9. It is clear from Fig. 9 that Mn promoted catalyst showed production of heavy hydrocarbon; thus, Mn aided in increasing the chain growth probability.

An attempt was made to correlate the catalyst characterization with catalyst performance. The catalyst property, such as ease of reducibility and the concentration of atomic Fe on the catalyst surface, was plotted against the CO conversion and selectivity (C_5+, CO_2 and CH_4) in Fig. 10. The ease of reducibility was defined based on the H_2 uptake during TPR analysis. Except nickel promoted catalyst, it can be observed from Fig. 10 that with an increase in H_2 uptake and percentage atomic Fe on the catalyst surface, there is an increase in CO conversion and C_5+ selectivity and a decrease in CO_2 and CH_4 selectivity.

Therefore, this work concludes that among Ni, Mg, Mn and Co promoters, the Mn promoted KCuFe catalyst supported on mesoporous alumina has shown better performance in terms of CO conversion and C_5+ selectivity. Further, the catalyst performance is well related to its characterization.

4. Conclusions

In this work, the effect of 1.0 wt % Ni, Mg, Mn and Co on KCuFe/mAl_2O_3 catalyst were investigated. BET analysis confirmed that the incorporation of promoters did not show a significant difference in textural properties. H_2-TPR studies concluded that the addition of promotors had eased the reduction of iron oxide. The second reduction peak for iron oxide appeared at ~70 °C less than that appeared in unpromoted catalyst. The ease of reduction followed the order Ni ≈ Mn > Mg > Co. The XPS studies on oxide catalysts revealed the interaction of Ni and Co with alumina, which probably resulted in increasing CH_4 selectivity. The atomic percentage of Fe on the surface of the catalyst was highest in the Mn-promoted catalyst, whereas Mg-promoted catalyst had the least iron on the surface, which is nearly 40 % less than that of Fe present on the surface of Mn-promoted catalyst. The Mn 2p XPS spectra of the oxide catalysts and the Mn K-edge XANES of reduced catalyst indicate a mixed Fe-Mn phase. The XANES analysis of Fe K-edge of all reduced promoted catalysts concluded that the extent of Fe reduction followed the order Ni- promoted > Mn-promoted > Mg-promoted ~ Co-promoted. The Fe K-edge EXAFS spectra in R space for all reduced promoted catalysts confirmed that the radial distance to first, second, the third shell is different due to the presence of different promotors. Therefore, the characterization studies had confirmed that the different promotors had interacted differently and consequently impacted FTS activity.

The FTS reaction was carried out at 270 °C, 300 psi, 2000 h^{-1} and H_2/CO = 1.25. Ni- promoted catalyst yielded high methane and low CO conversion, attributed to the inherent property of Ni for methane formation. The Co-promoted catalyst didn't affect the CO conversion, but it showed a decline in Olefins/Paraffins (O/P) ratio and a ~5% increase in methane selectivity. Mg-promoted catalyst performed better than Ni and Co-promoted catalysts. It helped in lowering the WGS activity, as evident from the marginal decline in CO_2 selectivity. Mn was the best promoter as it reduced the CH_4 and CO_2 selectivity by 2% and 5%, respectively, and increased the C_5+ selectivity from 77 % to 81 %. The increase in iron dispersion, as observed in XPS analyses, ease in reducibility as seen in H_2-TPR and XANES analysis, and the interaction of Mn and Fe at the electronic level are the possible reasons for the improved FTS activity and selectivity of Mn-promoted catalyst. Therefore, Mn-KCuFe/mAl_2O_3 is a potential catalyst with tolerance to impurities (CO_2 and CH_4) for commercial FTS process using syngas having low (~1.25) H_2/CO ratio.

CRediT authorship contribution statement

Sandeep Badoga: Conceptualization, Methodology, Writing - original draft. **Girish Kamath:** Investigation, Validation. **Ajay Dalai:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are thankful to Dr. Yongfeng Hu and Dr. Mohsen Shakouri for helping with XANES measurements at the Canadian Light Source (CLS), Saskatoon, Saskatchewan, a national research facility. The authors are sincerely thankful to the Saskatchewan Structural Sciences Centre (SSSC) at the University of Saskatchewan for the use of their XPS facility. The authors also acknowledge the financial support from the Natural Science and Engineering Research Council (NSERC) and Canada Research Chair (CRC) program.

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